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A two-year assessment of particulate air pollution and sources in Kuwait☆

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ABSTRACT

Background: Kuwait and the Gulf region have a desert, hyper-arid and hot climate that makes outdoor air sampling challenging. The region is also affected by intense dust storms. Monitoring challenges from the harsh climate have limited data needed to inform appropriate regulatory actions to address air pollution in the region.

Objectives: To compare gravimetric measurements with existing networks that rely on beta-attenuation measurements in a desert climate; determine the annual levels of $PM_{2.5}$ and PM_{10} over a two-year period in Kuwait; assess compliance with air quality standards; and identify and quantify $PM_{2.5}$ sources.

Methods: We custom-designed particle samplers that can withstand large quantities of dust without their inlet becoming overloaded. The samplers were placed in two populated residential locations, one in Kuwait City and another near industrial and petrochemical facilities in Ali Sabah Al-Salem (ASAS) to collect $PM_{2.5}$ and PM_{10} samples for mass and elemental analysis. We used positive matrix factorization to identify $PM_{2.5}$ sources and apportion their contributions.

Results: We collected 2339 samples during the period October 2017 through October 2019. The beta-attenuation method in measuring $PM_{2.5}$ consistently exceeded gravimetric measurements, especially during dust events. The annual levels for $PM_{2.5}$ in Kuwait City and ASAS were 41.6 ± 29.0 and 47.5 ± 27.6 $\mu g/m^3$, respectively. Annual $PM_{2.5}$ levels in Kuwait were nearly four times higher than the U.S. National Ambient Air Quality Standard. Regional pollution was a major contributor to $PM_{2.5}$ levels in both locations accounting for 44% in Kuwait City and 46% in ASAS. Dust storms and re-suspended road dust were the second and third largest contributors to $PM_{2.5}$, respectively.

Conclusions: The premise that frequent and extreme dust storms make air quality regulation futile is dubious. In this comprehensive particulate pollution analysis, we show that the sizeable regional anthropogenic particulate sources warrant national and regional mitigation strategies to ensure compliance with air quality standards.

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1. Introduction

Kuwait is a small country located at the northeastern corner of the Arabian Peninsula. The country is known for its desert and hyper-arid and hot climate. Because of its unique climate, two environmental exposures are amplified in this part of the world: dust storms and extreme hot temperatures (Alahmad et al., 2020; Li et al., 2020; Merlone et al., 2019). The northern parts of the Arabian Peninsula receive annually about 373 tons per km^2 of deposited dust (Al-Dousari et al., 2017). Kuwait city has 270 tons/km^2/year of dust deposited on it, one of the largest totals of any city in the world (Al-Dousari et al., 2013; United Nations Environment Programme, 2016). Ambient air temperatures in Kuwait are soaring to unprecedented levels (Alahmad et al., 2019). In 2016, the temperature reached 54 °C in Mitribah, Kuwait, which was the highest, officially recognized temperature to have been recorded in the last 76 years (Merlone et al., 2019). These extreme conditions make air quality measurement difficult. Conventional sampling devices are equipped with impaction substrates that are not suitable for harsh dust storms. Sampler inlets can clog and the substrates are prone to particle bounce, which reduces sampler accuracy (National Research Council, 2010). Furthermore, extremely hot outdoor temperatures may limit the ability to conduct fieldwork.

Previous air quality sampling studies in Kuwait measured air quality for 1 year between 2004 and 2005 (Alolayan et al., 2013; Brown et al., 2008). Al-Hemoud et al. utilized real-time $PM_{2.5}$ measurements from national networks (Al-Hemoud et al., 2019). These studies identified that Kuwait and Gulf countries have high levels of particulate air pollution from local and regional anthropogenic sources and, primarily, natural sources (from desert and locally generated dust). The annual $PM_{2.5}$ and PM_{10} concentrations in Kuwait were found to exceed the levels recommended by the World Health Organization (WHO) and the U.S. National Ambient Air Quality Standards (NAAQS) recommended levels. The Kuwait Environmental Public Authority (KEPA) does not have an annual standard for $PM_{2.5}$ or PM_{10}. The 24-h $PM_{2.5}$ and PM_{10} concentration limits were set in 2017 at 75 and 350 μg/m^3, respectively, based on the 99th percentile of maximum concentration over 1 year (Kuwait Environmental Public Authority, 2017). The previous sampling and speciation of particulate matter in Kuwait (conducted more than a decade ago) was limited in its temporal resolution as speciated samples were not obtained daily in the Kuwait City site and were obtained every 6th day in other background locations (Brown et al., 2008). The sparse literature on outdoor air pollution along with the challenging nature of air quality sampling in Kuwait makes it difficult to draw definitive inferences about air quality in Kuwait and the region. Meanwhile, local authorities in Kuwait lack critical new information that can guide many decisions about air quality standards and associated interventions.

We deployed custom-designed particle samplers to collect samples under severe dust storm episodes in Kuwait. We collected $PM_{2.5}$ and PM_{10} daily samples in two residential locations, one in Kuwait City and the other at a southern location in Ali Sabah Al-Salem (ASAS) near industrial and petrochemical facilities. After 15 years from the last sampling program, the aims of this new comprehensive particulate air quality assessment are to: 1) compare our gravimetric measurements with an existing network that relies on beta-attenuation measurements; 2) determine the annual levels of $PM_{2.5}$ and PM_{10}; 3) assess compliance to national and international air quality standards, and; 4) identify and quantify the sources of $PM_{2.5}$ in Kuwait.

2. Methods

2.1. Sampling locations

We placed particle samplers in two locations in Kuwait (Fig. 1; *Supplemental Figure A*). The first monitoring site, a rooftop of the Kuwait University College of Engineering Campus in Al-Khaldiya (29.3217° N, 47.9719° E), which is an urban residential area at the heart of Kuwait City (referred to as 'Kuwait City' onwards). It is less than 3 km away from the city's downtown. The second location, a rooftop of a local healthcare center in the Ali Sabah Al-Salem area (formerly known as Um Al-Haiman) (28.9570° N, 48.1548° E). This is a residential area located toward the south, about 50 km from Kuwait City (referred to as 'ASAS' onwards). This area is surrounded by oil fields, industrial operation facilities and a sewage treatment plant. The elevations at Kuwait City and ASAS sites were nearly 18 and 26 m above sea level, respectively. We collected daily $PM_{2.5}$ and PM_{10} at both sites during the period of October 2017 to October 2019.

2.2. Sampling and analyses

We employed custom-designed Harvard Impactor (HI) samplers that can collect large quantities of particles, especially during dust storm episodes. These HI samplers were developed by our team at the Harvard T. H. Chan School of Public Health and have previously been successfully field-tested in Kuwait (Brown et al., 2008). The sampler uses a slit acceleration jet and polyurethane foam (PUF) as an impaction substrate. This substrate is designed to have a high capacity to remove and withhold amounts of particles larger than the cut-points (10 and 2.5 μm) during dust storm events (Demokritou et al., 2004). The impaction surface of conventional devices can be overloaded during severe dust storm episodes; therefore, they are prone to particle bounce, affecting the sampler accuracy considerably (National Research Council, 2010). The capacity of our PUF impaction surface is 35 mg of PM, which makes it possible to collect particles in challenging desert environments (*Supplemental Figure B*).

$PM_{2.5}$ and PM_{10} filter samples were collected on a pre-weighed Teflon filter from 9:00 a.m. to 9:00 a.m. of the following day at a flow rate of 10 L/min. After collection, the filter samples were stored at ~4 °C before being shipped to the Harvard T.H. Chan School of Public Health for gravimetric analysis in a temperature- and humidity-controlled room. Coarse particles ($PM_{2.5\text{-}10}$) were calculated by subtracting the PM_{10} concentrations from the $PM_{2.5}$ concentrations. Energy Dispersive X-Ray Fluorescence (EDXRF; Epsilon5, PANalytical, The Netherlands) was used to measure trace elements. Black carbon (BC) was analyzed from the Teflon filters using an optical transmissometer (OT21, Magee Scientific) at 880 nm. For a subset of days, we also collected and analyzed quartz filters (47-mm prefired, Sunset Laboratories) for elemental carbon (EC) and organic carbon (OC) using a Thermo/Optical EC/OC Analyzer (Sunset Laboratories, Model 3) following the National Institute of Occupational Safety and Health (NIOSH 5040) protocol (Chow et al., 2001).

2.3. Quality assurance

All sampling and laboratory analysis protocols were developed by the Harvard T. H. Chan School of Public Health. Field/Sampling protocols were made available to all partners and were used to train

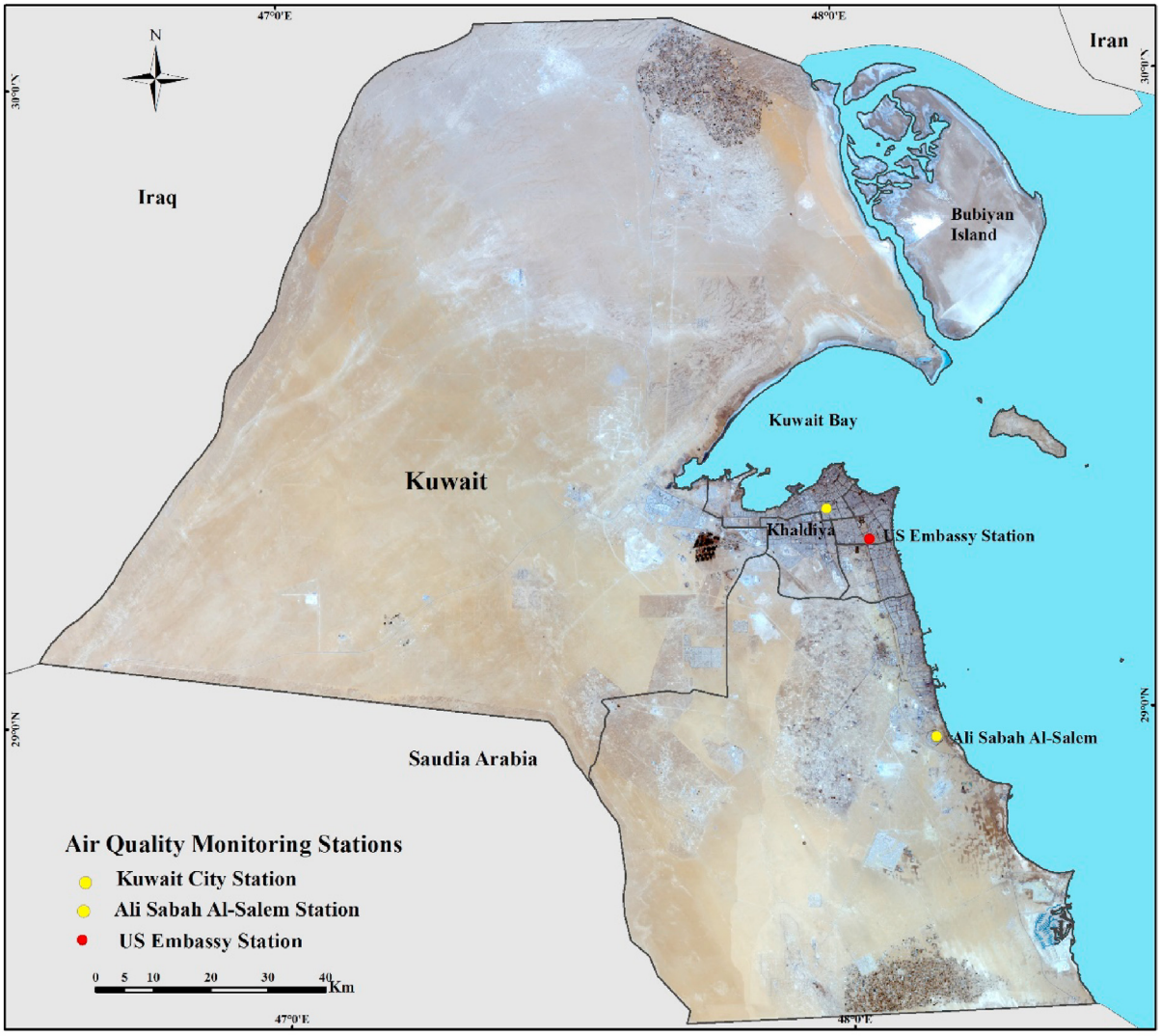


Fig. 1. Map of the two Harvard Impactor sampling sites in Al-Khaldiya (Kuwait City) and Ali Sabah Al-Salem residential areas (yellow points). The U.S. Embassy site is to the south of our Kuwait City site (red point). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

field technicians. Duplicate samples were collected using collocated samplers. These extra filters were used to assess measurement precision. The mass and crustal species of matched-pairs of collocated samples were visually examined for agreement using scatter plots. All mass measurements were blank-corrected using the mean mass of the field blanks for each size fraction and location. Limits of detection (LODs) for mass was calculated as 3 times the standard deviation of the field blank values. Elemental concentrations were reported only for selected elements, for which more than half of all observations were above the laboratory LODs: Na, Mg, Al, Si, S, Cl, K, Ca, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Br, Sr, Zr and Pb.

For each day when collocated samples were collected, we took the average measurement of the two samples. Samples were removed from the analysis when power failure was detected in the devices ($n = 2$), when $PM_{2.5}$ was higher than PM_{10} in the same day and same location ($n = 13$), and when $PM_{2.5}$ BC was higher than $PM_{2.5}$ mass in the same day and same location ($n = 1$). In December 2018, three observations in Kuwait City had exceptionally high levels of PM on days without recorded dust events. Independent measurements of visibility for these days did not correspond to typical dust storm values and neither Fe nor Si concentrations (markers for dust) were consistent with these high PM levels. Therefore, we removed these three observations from our analysis. We encountered consecutive missing days between June and July of 2018 and from January to July 2019 sampling occurred only every other day due to logistic restrictions.

2.4. Comparison datasets

In a secondary analysis, we also compared our gravimetric Harvard Impactor $PM_{2.5}$ concentrations against an existing monitoring network in Kuwait in order to assess the accuracy of the currently available information for regulators. The U.S. Embassy in Kuwait City has a monitoring site that uses the beta attenuation mass (BAM) method to measure and log the concentration of ambient particulate matter every hour (location is shown in Fig. 1). We matched our daily $PM_{2.5}$ in Kuwait City to the hourly $PM_{2.5}$ dataset from the U.S. Embassy (available: https://www.airnow.gov/international/us-embassies-and-consulates/#Kuwait$Kuwait_City). We calculated 24-h averages using hourly $PM_{2.5}$ values that correspond to our 9:00 a.m. to 9:00 a.m. sampling schedule. Additionally, we extracted $PM_{2.5}$ data from 33 other U.S. Embassies

located in different cities around the world. We calculated a $PM_{2.5}$ annual average for each city in 2019 and compared it to Kuwait City levels.

2.5. Dust event identification

To assign dust and non-dust days we used definitions that conforms with the World Meteorological Organization (WMO) codes for dust storms based on wind speed and horizontal visibility. Meteorological data was collected from the Kuwait Meteorological Department at the Kuwait International Airport. Three classifications of dust events were applied: suspended dust (dust haze), rising dust (blowing dust), and dust storm. Suspended dust event occurs when there is calm wind speed of 6–26 km/h and horizontal visibility in the range of 1–5 km (Al-Hemoud et al., 2018). Rising dust was defined as elevated dust occurring when winds are moderately active at a speed of 15–40 km/h and visibility is 1000 m or more (Al-Hemoud et al., 2018). A dust storm was defined as having surface winds exceeding 33 km/h, raising large quantities of dust into the air and reducing visibility to less than 1000 m (Al-Hemoud et al., 2018; McTainsh and Pitblado, 1987; World Meteorological Organization, 2005). Multiple dust categories could occur within one day. We classified dust event days as any day during which conditions met criteria for any of the three dust categories.

2.6. Source apportionment

Positive Matrix Factorization (PMF) – PMF was used to identify major sources of $PM_{2.5}$ and characterize the magnitude of their contribution to the total mass. PMF is a commonly used tool of receptor-models analyses (Hopke, 2003; Kim et al., 2003).

For species j in sample i, the predicted concentration (x_{ij}) is calculated using equation (1). However, both g_{ik} (the contribution of the kth factor to sample i) and f_{kj} (mass of species j per total mass of factor k) are unknown, and hence a regression analysis cannot be used. Instead, the PMF model applies a principle component analysis that relies on an eigenvector value of the correlation matrix, in which the residuals from predicting each data point are weighted individually by their uncertainty. PMF allows only non-negative loadings and calculates factor contributions and profiles by minimizing the objective function Q, where e_{ij} is the residual error from predicting the concentration, and σ_{ij} is the uncertainty in measuring the concentration, as shown in equation (2).

$$x_{ij} = \sum_{k=1}^{p} g_{ik} f_{kj} + e_{ij} \quad (1)$$

$$Q = \sum_{i=1}^{n} \sum_{j=1}^{m} \left(\frac{e_{ij}}{\sigma_{ij}}\right)^2 \quad (2)$$

The method for calculating the element concentration uncertainties was described elsewhere (Carrion-Matta et al., 2019; Polissar et al., 1998; Reff et al., 2007). Uncertainties in $PM_{2.5}$ and BC were pre-set at 10% for each. Using the signal-to-noise (S/N) ratio in PMF, species with S/N ratios below 2 were down-weighted (weak species).

Model Choice – We iteratively ran PMF models for $PM_{2.5}$, with each model run *a priori* including 3 to 5 factors. We chose the best model solutions based on interpretability and goodness of fit. Interpretability was determined by fingerprint elements in each factor source. Goodness of fit was determined by subtracting the measured concentration (x_i) from the predicted concentration ($\hat{x}_i$) and getting the percentage mean relative error (MRE), as in equation (3) (Carrion-Matta et al., 2019; Masri et al., 2015):

$$MRE = \left|\frac{x_i - \hat{x}_i}{x_i}\right| \times 100 \quad (3)$$

2.7. Data analysis

We stratified the average concentrations over PM size fractions, locations, months and dust event days. For each stratum, we reported the arithmetic mean and the standard deviation. In addition, we reported the overall annual mean by averaging all days during the study period. In a sensitivity analysis, to assess the effects of seasonality or days of the week on the overall annual levels of PM, we fitted a generalized linear model to calculate the annual concentrations (dependent variables) adjusting for calendar year, month and day of the week. Because these models adjust for month of the year, they provide a better estimate of annual concentrations than just simply averaging all samples. Any potential effect of months that were sampled more than others is adjusted for in the regression models. We used the U.S. Environmental Protection Agency (EPA) PMF model software (version 5.0) to perform the source apportionment analysis (Environmental Protection Agency, 2014). All other analyses were done using R (version 3.6.0).

3. Results

We collected 2339 samples between October 2017 and October 2019 (608 $PM_{2.5}$ and 596 PM_{10} samples in Kuwait City and 578 $PM_{2.5}$ and 557 PM_{10} in ASAS). About 37% of days were recorded to be dust event days.

We collected 141 matched pairs of collocated samples for quality assurance purposes. Comparing matched pairs in 1:1 reference line plot, there was a near perfect agreement for collocated $PM_{2.5}$ samples in both locations, even during days with dust events (Fig. 2). There was more scatter for the PM_{10} samples especially during dust event days. Collocated samples of crustal elements such as Si and Fe also showed satisfactory agreement for fine particles but less precision for large particles (Supplemental Figure C).

3.1. U.S. Embassy data comparison

We obtained 607 matched pairs of U.S. Embassy $PM_{2.5}$ and the Harvard Impactor $PM_{2.5}$ samples from Kuwait City (Table 1). The $PM_{2.5}$ averages of the U.S. Embassy and Harvard measures were 53.1 ± 36.8 μg/m^3 and 41.6 ± 29.0 μg/m^3, respectively; an average difference of 11.5 μg/m^3. The differences were more pronounced in June and July, and in days with dust events. The discrepancies are especially noticeable in dust event days that have high levels of coarse particles ($PM_{2.5\text{-}10}$) (*Supplemental Figure D*). Across days, the two measurements provide a similar signal. However, the U.S. Embassy measures were consistently higher (Fig. 3). Monthly and daily trends showed similar findings (*Supplemental Figures E & F*). The correlation between our Kuwait City and ASAS Harvard Impactor sites was similar to the correlation between our Kuwait City Harvard Impactor site and the U.S. Embassy site ($r = 0.91$, $p < 0.001$) (*Supplemental Figure G*).

3.2. Levels

Table 2 summarizes the mean concentrations of Harvard Impactor PM across locations, months and dust events, while Fig. 4 shows the medians and interquartile ranges for $PM_{2.5}$ (PM_{10} in

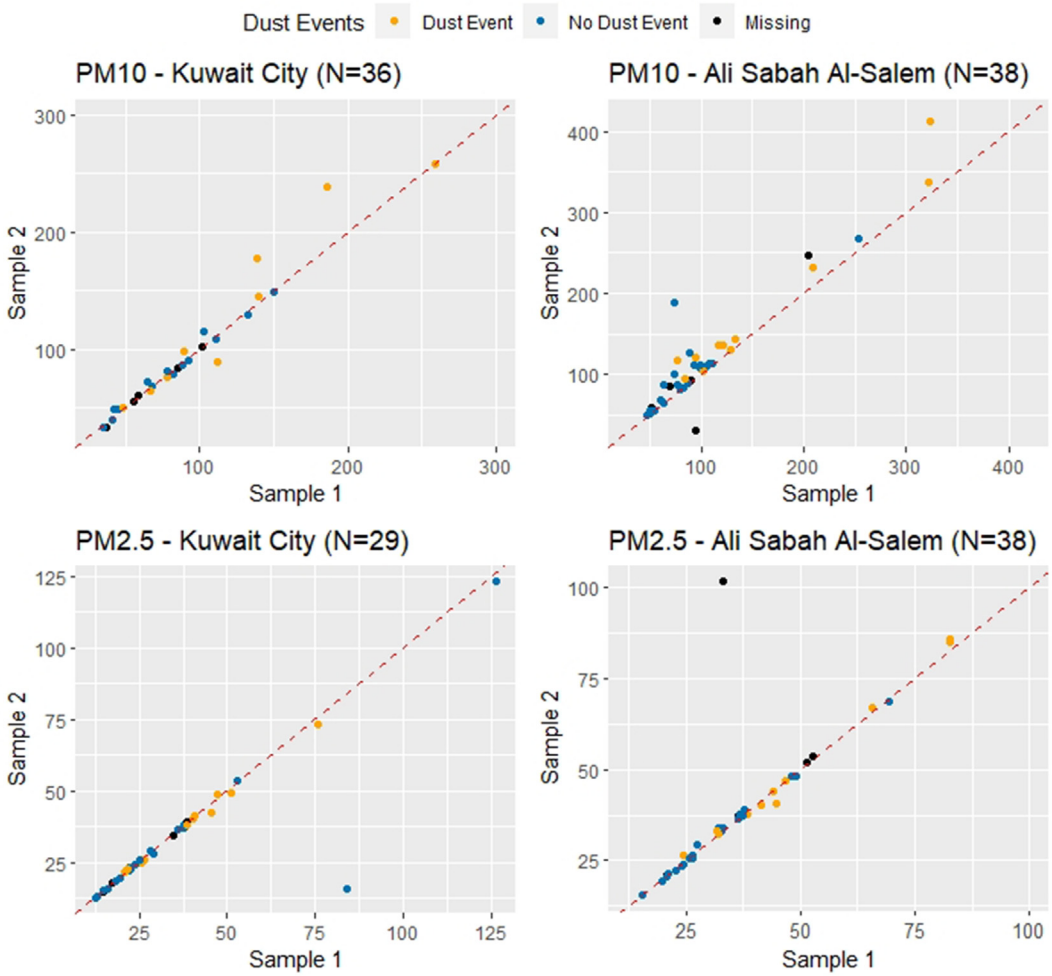


Fig. 2. Comparison of PM_{10} and $PM_{2.5}$ mass between matched pairs of collocate (duplicate) samples in Kuwait City and Ali Sabah Al-Salem (ASAS) with dust and no dust event days noted.

Supplemental Figure H). The overall mean of $PM_{2.5}$ in Kuwait City was lower than ASAS (41.6 ± 29.0 vs. 47.5 ± 27.6 μg/m^3). The overall PM_{10} mean in ASAS was 138.3 ± 79.5 μg/m^3, higher than Kuwait City; 112.4 ± 72.4 μg/m^3. There was considerable monthly and daily variability in PM concentrations, and ASAS had consistently higher concentrations than Kuwait City (daily variability in *Supplemental Figures I & J*). In the sensitivity analysis, after adjusting for day of the week, month and year, the regression estimation of annual PM levels did not meaningfully differ from the unadjusted averages reported in Table 2 (*Supplemental Table A*). The $PM_{2.5}/PM_{10}$ ratio was 0.38 (±0.12) and 0.37 (±0.11) in Kuwait City and ASAS, respectively. EC and OC were measured in 44 days in Kuwait City and 46 days in ASAS from January to April 2019. Mean concentrations of EC and OC were 3.4 and 3.6, respectively, in Kuwait City. The EC/OC ratio was about 0.9 in both locations with lower ratios observed during dust event days (*Supplemental Table E*; *Figure* L).

3.3. Compliance

We determined the number and percent of days where the $PM_{2.5}$ and PM_{10} levels exceeded the 24-h limits set by the WHO, USEPA and KEPA. Over the entire study period, $PM_{2.5}$ levels exceeded the daily WHO standard (25 μg/m^3) 74% of the time in Kuwait City and 90% in ASAS, while PM_{10} standards (50 μg/m^3) were exceeded in about 90% of days in both locations (Table 3). The U.S. NAAQS for $PM_{2.5}$ (35 μg/m^3) were exceeded in more than 51 and 67% of all days in our study period for Kuwait City and ASAS, respectively. KEPA $PM_{2.5}$ daily standards (75 μg/m^3) were exceeded in less than 10% of the days, while the PM_{10} standards (150 μg/m^3) were exceeded in only 3% of the days. We repeated the analysis looking at exceedance of standards restricted to data only in one year (2018) and it showed somewhat similar inferences (*Supplemental Table B*).

3.4. Source apportionment

Table 4 shows the average concentrations of species that were included in the PMF models across the two locations. We identified five source factors for $PM_{2.5}$ in both Kuwait City and ASAS. The error in predicting $PM_{2.5}$ was less than 2% for both locations. The identified five source factors in Kuwait City were: 1) sand and dust storms, 2) marine pollution, 3) regional pollution, 4) traffic and 5) road dust. The five sources in ASAS were: 1) sand and dust storms, 2) industrial emissions, 3) regional pollution, 4) traffic/other and 5) road dust (Table 5; *Supplemental Tables C & D*). Regional pollution was the dominant contributor in both locations accounting for 44% and 46% in Kuwait City and ASAS, respectively. Dust storms and re-suspended road dust were the second and third contributors, respectively. As expected, traffic emissions were higher in Kuwait City, while the industrial emissions were clearly seen in ASAS. The

Table 1
Comparison of matched pairs between Harvard Impactor (gravimetric) and U.S. Embassy (beta attenuation) $PM_{2.5}$ measurements between October 2017 and October 2019 in Kuwait City.

	Matched Pairs	Harvard $PM_{2.5}$	U.S. Embassy $PM_{2.5}$	Difference[a]	p-value[b]
		Mean ± SD (in μg/m^3)			
Overall	607	41.6 ± 29.0	53.1 ± 36.8	−11.5	<0.001
Year					
2017	66	44.8 ± 46.0	47.8 ± 49.1	−3.0	<0.001
2018	341	44.7 ± 28.9	55.8 ± 38.8	−11.1	<0.001
2019	200	35.4 ± 19.7	50.3 ± 27.1	−14.9	<0.001
Dust Events					
Dust Event	228	53.7 ± 33.0	70.6 ± 44.5	−16.9	<0.001
No Dust Event	370	34.5 ± 23.9	42.8 ± 26.5	−8.3	<0.001
Missing[c]	9	29.2 ± 9.8	34.7 ± 9.7	−5.5	0.001
Month					
1	49	40.3 ± 33.0	48.0 ± 42.0	−7.7	<0.001
2	40	39.5 ± 26.4	42.2 ± 24.6	−2.8	<0.001
3	45	34.5 ± 14.4	39.9 ± 15.1	−5.5	<0.001
4	44	38.1 ± 30.1	41.9 ± 34.1	−3.9	0.820
5	46	45.9 ± 22.3	56.1 ± 32.2	−10.3	0.321
6	44	50.6 ± 39.4	73.4 ± 57.0	−22.8	<0.001
7	39	48.2 ± 36.6	70.2 ± 50.4	−22.0	<0.001
8	57	37.9 ± 22.9	57.3 ± 28.7	−19.4	<0.001
9	60	42.3 ± 16.6	56.8 ± 17.8	−14.5	<0.001
10	66	54.8 ± 45.9	72.9 ± 51.1	−18.1	<0.001
11	62	31.2 ± 14.9	35.1 ± 15.2	−4.0	<0.001
12	55	36.9 ± 18.1	42.4 ± 18.3	−5.5	<0.001

[a] Calculated as average Harvard $PM_{2.5}$ minus U.S. Embassy $PM_{2.5}$.
[b] P-values calculated using a paired *t*-test with unequal variance of the log transformed $PM_{2.5}$.
[c] There were days with missing data on wind speed and visibility that we were not able to classify.

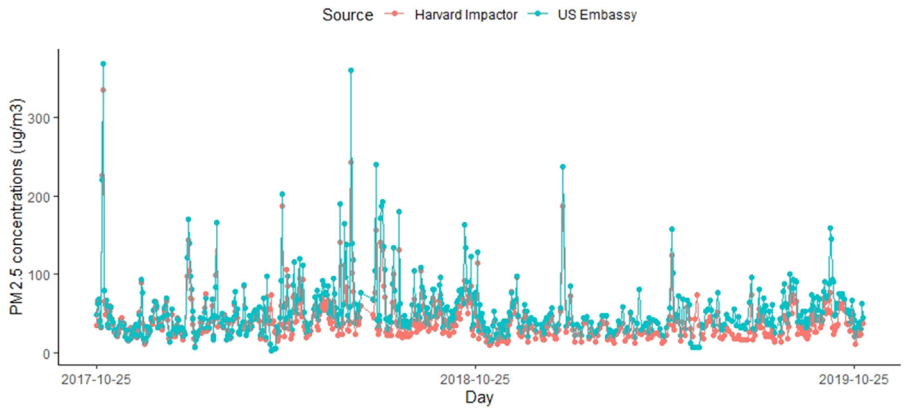


Fig. 3. Comparison between the U.S. Embassy (beta attenuation) and Harvard Impactor (gravimetric) daily $PM_{2.5}$ measurements during the study period in Kuwait City. The figure shows matched pairs for each day. Unit of measurement is μg/m^3.

interpretations for the outputs of the PMF models are provided in the supplemental material.

4. Discussion

This is the most comprehensive particulate air quality assessment in Kuwait to date. The unique air quality dataset created for Kuwait and the Gulf region was acquired with daily sampling of inhalable and fine particles at two sites over two users employing a robust sampling method that addressed challenges in collecting samples in a harsh desert environment.

We performed collocated sampling as a quality assurance test of the reliability of our mass, Si and Fe measurements. The best agreements were for the $PM_{2.5}$ mass parameters, with only a few apparent outliers. There was more scatter for the PM_{10} parameters. A possible explanation for the higher variability of the PM_{10} parameters is in the sampling conditions. We can presume that during high winds of dust storms the particles larger than 2.5 μm are not distributed uniformly in the air. There is also likely to be considerable turbulence, with strong micro-gradients of wind velocity. Taking these factors into consideration, it is reasonable to assume that air samples collected by the two collocated samplers could have different amounts of these larger particles. The better agreement for the $PM_{2.5}$ measurements can be a result of their uniform distribution in the air, even under dust storm conditions. We can conclude that the $PM_{2.5}$ measurements have higher reliability, and the PM_{10} measurements are relatively less reliable.

While the gravimetric and real-time BAM measurements of $PM_{2.5}$ provided similar trends and associations, we showed that the BAM measurements consistently exceed gravimetric measurements, especially during dust events. Beta monitors rely on the relationship of β-ray attenuation to particle mass to determine the mass concentration of atmospheric aerosols (Chueinta and Hopke, 2001). They are useful for measuring short-term (e.g. hourly)

Table 2
$PM_{2.5}$ and PM_{10} levels (mean ± standard deviation in μg/m^3) at the two Harvard Impactor sampling locations (Kuwait City & Ali Sabah Al-Salem; ASAS).

	$PM_{2.5}$				PM_{10}			
	N	Kuwait City	N	ASAS	N	Kuwait City	N	ASAS
Overall	608	41.6 ± 29.0	578	47.5 ± 27.6	596	112.4 ± 72.4	557	138.3 ± 79.5
Dust Events								
Dust Event	228	53.7 ± 33.0	217	58.8 ± 31.3	221	152.3 ± 93.4	200	179.7 ± 100.4
No Dust Event	371	34.5 ± 23.8	353	40.7 ± 22.8	366	89.4 ± 41.9	349	115.3 ± 52.4
Missing[a]	9	29.2 ± 9.8	8	40.5 ± 11.9	9	72.7 ± 27.3	8	102.2 ± 53.5
Month								
1	49	40.3 ± 33.0	47	44.5 ± 25.1	48	130.6 ± 105.7	46	168.4 ± 120.4
2	40	39.5 ± 26.4	41	40.8 ± 18.2	41	105.2 ± 72.5	41	118.4 ± 73.8
3	45	34.5 ± 14.4	45	40.0 ± 16.0	45	114.9 ± 57.1	45	125.3 ± 72.4
4	44	38.1 ± 30.1	45	41.5 ± 30.9	41	122.0 ± 91.3	44	141.6 ± 101.5
5	46	45.9 ± 22.3	43	51.6 ± 28.4	43	146.3 ± 79.0	42	170.3 ± 109.9
6	44	50.6 ± 39.4	18	43.3 ± 15.9	42	127.7 ± 60.3	17	118.2 ± 39.6
7	39	48.2 ± 36.6	41	63.0 ± 42.7	36	126.0 ± 98.9	34	151.3 ± 54.1
8	57	37.9 ± 22.9	59	50.7 ± 25.3	59	101.2 ± 63.3	54	146.1 ± 51.9
9	60	42.3 ± 16.6	60	52.3 ± 16.4	60	101.4 ± 40.2	60	149.2 ± 50.0
10	66	54.8 ± 45.9	65	56.3 ± 42.7	63	122.7 ± 70.9	58	146.0 ± 92.0
11	62	31.2 ± 14.9	57	36.8 ± 15.6	62	76.0 ± 56.1	57	98.5 ± 61.7
12	56	36.9 ± 18.0	57	45.0 ± 18.8	56	99.5 ± 47.7	59	123.9 ± 47.8

[a] There were days with missing data on wind speed and visibility that we were not able to classify.

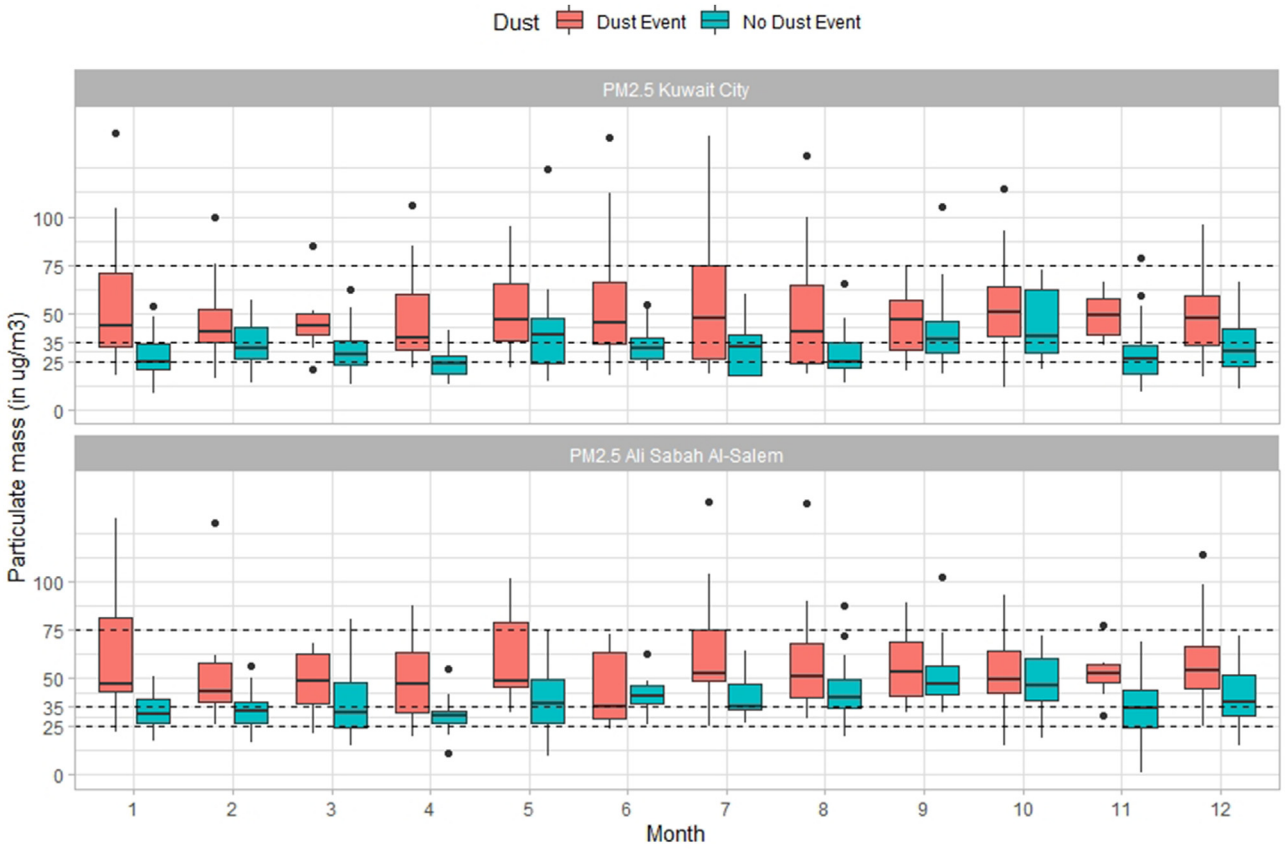


Fig. 4. Box plots of $PM_{2.5}$ measurements (in μg/m^3) stratified by months, dust events, and locations. Dotted lines represent Kuwait Environmental Public Authority (KEPA; 75 μg/m^3), U.S. National Ambient Air Quality Standards (NAAQS; 35 μg/m^3) and World Health Organization (WHO; 25 μg/m^3) 24-h standards.

temporal variation in PM concentrations (Takahashi et al., 2008). On the other hand, gravimetric measurements, which are the U.S. EPA standard method, need longer sample durations to accurately determine the mass of atmospheric particles collected on filters, yielding time-weighted averages, typically for 24-h. In BAMs, the quantitative attenuation of the β-rays passing through the sample depends on the composition of $PM_{2.5}$ that corresponds to the absorption of these rays by the particles collected on the filter. However, different compositions of $PM_{2.5}$ can have different ratios of attenuation to mass. There are 'standard' ratios that are incorporated by the manufacturers into the BAM software, based on studies of collocated gravimetric samples. These ratios can be different, depending on the composition at the sampling location(s) chosen for the comparison with gravimetric measurements. It is

Table 3
Number and percentage of days of study where particulate matter concentrations exceeded the 24-hr standards from October 2017 to October 2019.

	24-hr Standard	Kuwait City	ASAS
$PM_{2.5}$			
WHO	25 μg/m3	447/608 (73.5%)	522/578 (90.3%)
NAAQS	35 μg/m3	311/608 (51.2%)	385/578 (66.6%)
KEPA	75 μg/m3	40/608 (6.6%)	49/578 (8.5%)
PM_{10}			
WHO	50 μg/m3	529/596 (88.8%)	537/557 (96.4%)
NAAQS	150 μg/m3	108/596 (18.1%)	164/557 (29.4%)
KEPA	350 μg/m3	15/596 (2.5%)	15/557 (2.7%)

ASAS; Ali Sabah Al-Salem, WHO; World Health Organization, NAAQS; National Ambient Air Quality Standards (US), KEPA; Kuwait Environmental Public Authority.

Table 4
Mean concentrations of species (in $\mu g/m^3$) in the two Harvard Impactor sampling locations (Kuwait City and Ali Sabah Al-Salem; ASAS).

	$PM_{2.5}$[a]		PM_{10}	
	Kuwait City	ASAS	Kuwait City	ASAS
BC	2.865	2.963	3.471	3.772
Na	0.458	0.473	0.688	0.653
Mg	0.512	0.590	1.480	1.858
Al	1.299	1.510	3.816	4.506
Si	2.736	3.465	9.017	11.287
S	3.970	4.290	4.095	4.311
Cl	0.088	0.165	0.552	0.717
K	0.457	0.736	1.123	1.558
Ca	1.765	2.143	9.782	12.957
Ti	0.080	0.086	0.294	0.357
V	0.013	0.016	0.021	0.027
Cr	0.003	0.004	0.015	0.020
Mn	0.021	0.035	0.073	0.110
Fe	1.079	1.205	4.019	5.092
Co	0.002	0.002	0.008	0.010
Ni	0.006	0.007	0.015	0.020
Cu	0.013	0.013	0.033	0.025
Zn	0.050	0.238	0.086	0.344
Br	0.018	0.019	0.025	0.029
Sr	0.012	0.013	0.052	0.070
Zr	0.004	0.003	0.015	0.013
Pb	0.014	0.028	0.018	0.039

Values represent concentration of species over the entire study period (October 2017 to October 2019).
[a] Only $PM_{2.5}$ samples were used in the PMF models.

Table 5
Source contributions of $PM_{2.5}$ in the two locations in Kuwait (Kuwait City and Ali Sabah Al-Salem; ASAS).

$PM_{2.5}$ Source Factors	Kuwait City	
	$\mu g/m^3$	%
Regional pollution	17.9	43.6
Sand and dust storms	8.2	19.9
Road dust	7.2	17.5
Traffic	6.6	16.0
Marine pollution	1.2	3.0
$PM_{2.5}$ Source Factors	**ASAS**	
	$\mu g/m^3$	%
Regional pollution	21.6	46.4
Sand and dust storms	13.4	28.8
Road dust	5.3	11.4
Traffic/Other	3.9	8.4
Industrial emissions	2.3	5.0

possible that these standard ratios may not reflect the desert environment in Kuwait. Thus, the overall higher concentrations of BAM measurements we found could possibly be explained by the BAM using an inappropriate ratio. Additionally, the size-selective inlet used for BAM is subject to bounce-off if it is overloaded by extreme dust events. This bounce-off would result in collecting artifact particles on the BAM filter, resulting in higher measurements. The dataset we produced from our gravimetric sampling can be used to statistically calibrate the BAM measurements in the region (Yanosky et al., 2002). Moreover, if new inlets are designed and validated to overcome particle bounce, they can be used to retrofit BAMs in the dust storm regions. Given the strong correlations between our Kuwait City site and the further away ASAS site, the large discrepancies in the concentrations we observed with the U.S. Embassy in Kuwait City are not likely due to spatial differences in air pollution levels.

The annual $PM_{2.5}$ means in Kuwait City and ASAS were 41.6 $\mu g/m^3$ and 47.5 $\mu g/m^3$, respectively, nearly four times higher than the U.S. NAAQS (12 $\mu g/m^3$), and nearly five times higher than the WHO standard (10 $\mu g/m^3$). Our findings showed higher concentrations compared to the previous sampling campaign more than a decade ago in which the $PM_{2.5}$ levels were 38 and 37 $\mu g/m^3$ at similar sites, respectively (Brown et al., 2008). Similarly, the corresponding annual PM_{10} means were 112.4 and 138.3 $\mu g/m^3$, respectively, which is about six times higher than the WHO standard (20 $\mu g/m^3$) and higher than the measured concentrations in 2004–2005; 92.8 and 90.4 $\mu g/m^3$, for Kuwait City and ASAS, respectively (Brown et al., 2008). The differences between the two residential locations are understandable given that the southern area is surrounded by oil fields, industrial operation facilities and a sewage treatment plant. Yet, the source apportionment analysis showed that regional anthropogenic and dust pollution sources dominate both locations in a similar pattern, as compared to local traffic and industrial emissions.

Kuwait's overall $PM_{2.5}$ levels are considerably greater than levels observed in Europe and the U.S. For example, country-wide annual levels of $PM_{2.5}$ in the U.S, Spain, Switzerland or Greece are estimated to be approximately 12, 11, 18 and 22 $\mu g/m^3$, respectively (Liu et al., 2019). However, we compared Kuwait City's $PM_{2.5}$ levels with 33 other cities in the developing world using the available data from the U.S. Embassies in 2019 (*Supplemental Figure K*). Kuwait City was considerably lower than South Asian capital cities such as New Delhi, Dhaka or Kabul. Moreover, other cities in the Gulf (Manama, Dubai and Dhahran) appear to be more or less similar to Kuwait City.

The $PM_{2.5}/PM_{10}$ ratio was 0.38 and 0.37 in Kuwait City and ASAS, respectively. These ratios vary geographically around the world. In North America the ratio is around 0.60 (Dockery and Pope, 1994), in Central and Eastern Europe between 0.55 and 0.78 (Houthuijs et al., 2001), and less than 0.50 in Eastern Asia (Hopke et al., 2008). In deserts such as Coachella Valley, California, U.S., the ratio was around 0.45 (Ostro et al., 2000), while in Jeddah, Saudi Arabia, the ratio was 0.20 (Lim et al., 2018). In Kuwait, the previous sampling program between 2004 and 2005 showed the ratio to be around 0.47 (Alolayan et al., 2013; Brown et al., 2008). Compared to the current ratio (decreasing by about 10 percentage points), it could indicate an increase in large and coarse particles that is likely attributable to an increase in the frequency of dust storms in the region over the past decade (Li et al., 2020). According to the dust classification by the Kuwait Meteorological Department, about 37% of days were recorded to be dust event days. It is prudent; however, to assume that even days that are classified as no dust events may also have a sizable amount of resuspended dust contribute to their average PM concentrations. The current KEPA 24-h limits for $PM_{2.5}$ and PM_{10} are 75 and 350 $\mu g/m^3$, respectively and were exceeded in less than 10% of the days. Such high regulatory limits (compared to other international standards) may help national compliance but limit public health and economic gains from pollution reduction.

Re-evaluation of the regulatory standards is warranted in a cost-benefit framework that considers economic, health and equity concerns. The stringent WHO standards do not seem to be within reach for Kuwait, while the U.S. NAAQS can represent a more realistic interim target of pollution standards for regulators. Meanwhile, more regional studies are needed to develop an air quality standard that reflect the regional characteristics of particle composition as well as the epidemiological adverse health effects of air pollution on the underlying populations. Although exposure to particulate air pollution has been associated with adverse health effects in studies conducted worldwide (Liu et al., 2019), effects of pollutants specific to a region (such as high dust levels) will require region specific studies.

During dust event days across many months, the entire interquartile range of $PM_{2.5}$ was above the NAAQS and WHO 24-h limits in both locations. These dust events cannot be prevented which leaves secondary prevention strategies such as keeping homes tight during dust storms, as the mainstay of dust exposure prevention (Yuan et al., 2020). Vegetation may be helpful in lowering $PM_{2.5}$ levels in Kuwait as seen in the effect of the green-tree belt that lies above Al-Jahra metropolitan area in Kuwait (Al-Hemoud et al., 2019). Sustainable rehabilitation and restoration of "hot spot" desert areas can mitigate the transboundary dust events affecting Kuwait (Al-Hemoud et al., 2020). Additionally, street sweeping practices that minimize aerosolization of dust, especially when coupled with street washing, have the potential to mitigate PM emission from road dust resuspension (AIRUSE LIFE, 2013; Amato et al., 2016).

The premise that frequent and extreme dust storms in Kuwait make air quality improvements futile is dubious. The anthropogenic $PM_{2.5}$ in Kuwait City and ASAS, as determined by the PMF analysis, is quite substantial. This is consistent with the 2005 source apportionment study in Kuwait that identified oil combustion, the petrochemical industry and other anthropogenic sources transported from outside the country as 35% of total $PM_{2.5}$ (Alolayan et al., 2013). This leaves an opportunity to intervene on anthropogenic sources and setting a stricter standard for traffic and industrial emissions. However, local measures, such as better vehicular fuel economy, traffic regulations and establishing industrial emissions inventory may not be sufficient on their own. They will need to be joined with regional actions to achieve significant reductions in air pollution and public health gains. It is unlikely that all these anthropogenic sources originate from within Kuwait, simply because of its small size and the geographical proximity to oil producing neighboring countries (Iraq, Saudi Arabia and Iran).

This study has a number of limitations. First, we had missing days between June and July of 2018 and sampled only every other day from January to July 2019. This may have affected our annual average concentrations since summer months (June and July) have high rates of dust storms (Li et al., 2020). However, we fitted a generalized linear regression model adjusting for year, month and day of the week, and we did not see an appreciable change in annual PM concentration averages. Second, in classifying dust event days we did not take into account the severity of dust storms. Additionally, carbon species may exhibit seasonal variations and can be affected by dust (Zhang et al., 2017). We were not able to investigate the effect of dust storms and seasonality on carbon species due to the limited amount of EC and OC samples we collected. Finally, PMF relies on correlation patterns between speciated elements rather than *a priori* knowledge about the chemical profiles of sources to generate source contribution estimates. This leaves a room for subjective interpretability of source factors, and therefore should be treated with caution.

5. Conclusions

More than fifteen years since the last air sampling program, this comprehensive particulate air quality assessment in Kuwait generated an updated and unique dataset for the country and the whole Gulf region that can advance air quality regulations aimed at protecting public health. Using custom-designed samplers that can withstand the harsh nature of this desert climate, we showed that the existing monitoring networks (like the U.S. Embassy site) that rely on real-time beta-attenuation measurements of $PM_{2.5}$ chronically overestimate pollution levels, especially during dust event days. Because dust is ubiquitous, compliance to international pollution standards may be hard to reach, but there is still an ample space to improve especially when anthropogenic sources are tackled nationally and regionally. Due to the small geographic and population size compared to the neighboring countries, it may be difficult to see results when interventions are made unilaterally by Kuwait. Regional cooperation is; therefore, inevitable in order to achieve significant reductions in air pollution and subsequent public health gains.

Author contributions

Dr Alahmad had full access to the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. *Concept and design:* Barrak Alahmad, Petros Koutrakis, *Acquisition, analysis, or interpretation of data:* All authors. *Drafting of the manuscript:* Barrak Alahmad, Jack M. Wolfson. *Critical revision of the manuscript for important intellectual content:* All authors. *Statistical analysis:* Barrak Alahmad, *Supervision:* Eric Garshick, Aaron S. Bernstein, Joel Schwartz, Petros Koutrakis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2021.117016.

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